

06_Module_6_Advanced_SRE_Capstone

Module 6 — Advanced SRE & Capstone

Duration: 6 Hours

6.1 Kubernetes Reliability Patterns

Kubernetes does not make your system reliable by itself. It provides the primitives; you must configure them correctly.

6.1.1 PodDisruptionBudget (PDB)

A PDB limits how many pods can be voluntarily disrupted at a time. Without it, node maintenance or cluster autoscaling can kill all replicas simultaneously.

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: transfer-service-pdb
spec:
  minAvailable: 2           # Keep at least 2 pods running
  selector:
    matchLabels:
      app: transfer-service
```

Or using maxUnavailable:

```
spec:
  maxUnavailable: 1         # Never disrupt more than 1 pod at a time
```

Why it matters: Without a PDB, a node drain during an incident could kill all your transfer service pods simultaneously, causing downtime.

6.1.2 Topology Spread Constraints

Ensures pods are distributed across failure domains (nodes, zones, regions).

```
spec:
  topologySpreadConstraints:
    - maxSkew: 1
      topologyKey: topology.kubernetes.io/zone # Spread across zones
      whenUnsatisfiable: DoNotSchedule
      labelSelector:
        matchLabels:
          app: transfer-service
```

Why it matters: If all pods run in one availability zone and that zone fails, the service is completely down. Spread constraints ensure pods are distributed.

6.1.3 Priority Classes

Critical services should not be evicted to make room for less important ones.

```
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
  name: high-priority
value: 1000
globalDefault: false
description: "Critical banking services"
---
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
  name: low-priority
value: 100
globalDefault: false
description: "Batch jobs, non-critical workloads"
```

Why it matters: During resource contention, the scheduler evicts lower-priority pods first, protecting critical services.

6.1.4 Resource Requests and Limits

Without proper requests, pods can overcommit node resources and cause neighbour disruption.

```
spec:
  containers:
    - name: transfer-service
      resources:
        requests:      # Guaranteed minimum
          memory: "256Mi"
          cpu: "250m"
        limits:        # Hard cap
          memory: "512Mi"
          cpu: "500m"
```

6.1.5 Cluster Autoscaler

Automatically adds/removes nodes based on unschedulable pods.

```
# cluster-autoscaler config
autoDiscovery:
  clusterName: union-bank-production
maxNodeCount: 20
minNodeCount: 3
scaleDownEnabled: true
scaleDownDelayAfterAdd: 10m
```

6.1.6 Liveness, Readiness, and Startup Probes

Three probes serve different purposes:

Probe	Purpose	Failure Action
livenessProbe	Is the app alive? (not hung)	Restart the pod

Probe	Purpose	Failure Action
readinessProbe	Is the app ready to serve traffic?	Remove from Service
startupProbe	Has the app finished starting?	Delay liveness checks

```
spec:
  containers:
    - name: transfer-service
      livenessProbe:
        httpGet:
          path: /healthz
          port: 8080
        initialDelaySeconds: 30
        periodSeconds: 10
        failureThreshold: 3
      readinessProbe:
        httpGet:
          path: /ready
          port: 8080
        initialDelaySeconds: 5
        periodSeconds: 5
      startupProbe:
        httpGet:
          path: /startup
          port: 8080
        initialDelaySeconds: 0
        periodSeconds: 2
        failureThreshold: 30 # Allow 60 seconds to start
```

6.2 Capacity Planning

The Process

Step 1 — Measure current usage (90 days of data)

Database disk usage:

Jan 1:	1.2 TB
Feb 1:	1.35 TB
Mar 1:	1.5 TB

Growth: ~150 GB/month

Current capacity: 2 TB
 Remaining: 500 GB
 Time to full: $500 \div 150 = 3.3$ months

Step 2 — Project forward with forecasting

```
# Disk full prediction
predict_linear(
  node_filesystem_free_bytes{mountpoint="/data"}[30d],
  3600 * 24 * 365 # 1 year
) < 0
```

Step 3 — Act at threshold

Threshold	Action
< 50%	Normal operations

Threshold	Action
50–70%	Plan capacity increase
70–80%	Procure additional capacity
> 80%	Immediate action — this is an incident

Banking Capacity Events

Event	Impact	Preparation
Salary payment day (25th–27th)	3–5x ATM and transfer volume	Pre-scale 48h ahead
End of month	High transaction volume	Increase connection pools
Christmas / New Year	Peak ATM withdrawals	Ensure cash and network capacity
New product launch	Unpredictable spike	Load test before launch
CBN policy change	Compliance-driven surge	Coordinate with business

6.3 SRE for Multi-Region Deployments

For critical banking services, consider multi-region deployment:

Pattern	RTO	RPO	Cost
Active-Passive	5–15 min	5 min	2x
Active-Active	< 1 min	Near-zero	3x+
Cold Standby	1–4 hours	1 hour	1.5x

Key implementation considerations: - Data consistency across regions (synchronous vs asynchronous replication) - DNS-based traffic routing (route53, CloudDNS with health checks) - Regional PDBs to prevent cross-region disruption - Cross-region failover testing (at least quarterly)

6.4 AI-Driven Reliability

Current State

SRE Task	Traditional	AI-Assisted
Anomaly detection	Static thresholds	ML models learn normal patterns
Root cause analysis	Manual log search	AI correlates events across systems
Capacity forecasting	Linear extrapolation	ML with seasonality + growth curves
On-call routing	Fixed escalation	AI routes by past handling patterns
Post-mortem drafting	Manual from notes	AI drafts timeline from incident data

Practical Tools

- **Datadog AI** — anomaly detection, intelligent alerting
- **Dynatrace Davis AI** — automated root cause analysis
- **AWS DevOps Guru** — ML-powered operational recommendations
- **Claude / GPT-4** — runbook and post-mortem drafting

Important Caveat

AI tools are **assistants, not replacements**. An AI can suggest a root cause — a human must verify it. AI can draft a post-mortem — a human must review it. The judgment, context, and accountability remain with the engineer.

6.5 Capstone Project (2.5 Hours)

Brief

Your team has been hired as SRE consultants for Union Bank. The board has approved a reliability transformation initiative. One service has been assigned to your team.

Options: - **A:** ATM Network (nationwide, 200+ ATMs, 24/7 operations) - **B:** Mobile Banking App (1M+ users, fastest-growing channel) - **C:** Online Transfer Service (inter-bank, NIP/NIBSS integrations) - **D:** Internet Banking Portal (regulatory-sensitive, business customers)

Deliverables

1. Service Overview (10 min) - Define the service boundary - Identify all dependencies (internal and external) - Map the user journey - Quantify business impact of 1-hour downtime

2. SRE Framework (60 min)

Component	Requirements
SLIs	4+ SLIs with formulas and aggregation strategy
SLOs	Targets with justification and safety margin analysis
Error Budget	Monthly budget calculation + policy (GREEN/YELLOW/RED/EXHAUSTED)
Burn Rate Alerts	4 windows with thresholds and severity assignments
Monitoring	Prometheus metrics list (10+), Grafana dashboard design, log strategy
Runbooks	3 runbooks for critical alerts
Toil Elimination	3 toil items with automation plan and ROI estimate
On-Call	Rotation design for team of 8, escalation path
Chaos	2 Game Day experiments with hypothesis, blast radius, and success criteria
K8s	Deployment YAML with PDB, probes, spread constraints, priority class

3. Presentation (15 min per group) - 10 min: Present your framework - 5 min: Q&A from class

Scoring Criteria:

Criteria	Weight
SLIs/SLOs are specific and measurable	20%
Error budget policy is actionable	15%
Burn rate alerts are properly configured	15%
Runbooks are detailed and actionable	15%

Criteria	Weight
Toil elimination plan has real ROI	10%
Chaos experiments test meaningful failure modes	10%
K8s config follows reliability best practices	10%
Presentation clarity and defence of decisions	5%

6.6 Course Close — 30-Day Action Plan

Week 1

- ☐ Read Google SRE Book Chapters 1–4
- ☐ Identify 3 toil items in your team
- ☐ Propose one error budget to your team lead

Week 2

- ☐ Define one SLI/SLO for a service you manage
- ☐ Install Prometheus + Grafana in test environment
- ☐ Configure 1 burn rate alert

Week 3

- ☐ Write your first runbook
- ☐ Run one chaos experiment in staging
- ☐ Review one post-mortem from a past incident

Week 4

- ☐ Present your SRE framework to your engineering manager
 - ☐ Automate one toil item
 - ☐ Schedule your first Game Day
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Key Terms

Term	Definition
PodDisruptionBudget	Limits voluntary disruptions to pods during maintenance
Topology Spread	Distributes pods across failure domains (zones, regions)
Priority Class	Marks a pod's importance for scheduling and eviction decisions
Cluster Autoscaler	Automatically adds/removes nodes based on demand
Multi-Region	Deploying services across geographic regions for disaster recovery
RTO	Recovery Time Objective — max acceptable downtime
RPO	Recovery Point Objective — max acceptable data loss
Game Day	Structured chaos experiment with hypothesis and fix

Further Reading

- [Google SRE Book, Chapter 17 — Testing Reliability](#)
- [Google SRE Workbook, Chapter 10 — Production Readiness Review](#)
- [Kubernetes Production Patterns](#)
- [Chaos Engineering Principles](#)
- [Multi-Region Deployment Patterns — AWS](#)